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Thermo-viscoelastic laminate-based finite element modeling of fused filament fabrication direct 4D printing

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Abstract

Recent advances in fused filament fabrication direct 4D printing necessitate new, accurate modeling approaches to computationally predict the programmed shape change and temperature-dependent mechanical properties of direct 4D printed designs. In this paper, a thermo-viscoelastic, laminate-based finite element modeling approach is proposed to meet this need. The model enables the simulation of the deployment and the mechanical characterization of thin, direct 4D printed structures in one simulation. Unlike conventional modeling approaches based on thermal expansion alone, the model introduces a pre-strain in the material model during model initialization. This creates realistic shape memory behavior during deployment based on a modified, generalized Maxwell model and enables the simulation of subsequent shape memory cycles. The approach is validated using several benchmark examples, including a sequentially deploying, multi-material bilayer to demonstrate the generality of the method and a direct 4D printed wave spring to illustrate a potential application. The method can predict the activation temperatures of the sequentially deploying bilayer to within 2 °C of the mean experimental values and the wave spring's shape and response force to within one standard deviation of the mean experimental values. The predictive simulation of the time-temperature dependent, direct 4D printed shape changing behavior enables designers to gain new, significant insight into the structural performance of direct 4D printed designs and to systematically explore the design space as well as material and manufacturing process interactions.

Supplementary material for this article is available [online](#)

Keywords: direct 4D printing, finite element modeling, material modeling, multi-material additive manufacturing

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1. Introduction

4D printing can be defined as a targeted change of an additively manufactured (AM) structure to modify its shape, properties, and functionality over time [1]. 4D printing of shape memory polymers (SMPs), for example, can be used to fabricate novel structures, such as self-deploying structures [2, 3], morphing structures [4–6], and actuators [7, 8], and thus opens new possibilities for various applications. Contrary to traditional 4D printing, the SMP is programmed during fabrication using direct 4D printing, creating a single-step approach to the fabrication and programming of the desired shape and behavior change [2, 9, 10]. Direct 4D printing is often used to create thin structures that have bending, twisting, shearing and axial extension behavior using single or multimaterial bilayers, gradients and melt-blend approaches [11, 12]. In fused filament fabrication (FFF) direct 4D printing, the programmed shape change is embedded via the AM pre-strain and can be controlled with two main sets of decoupled parameters. The deformation mode is controlled by the AM direction and the AM layer arrangement [12, 13]. The deformation magnitude is controlled using process parameters, such as extrusion temperature and feedrate [12–15].

One of the key challenges in 4D printing is the design of structures that show repeatable and predictable behavior [16]. Recent advances in traditional 4D printing and direct 4D printing [2, 6–9, 17–20] have created a considerable need for accurate modeling approaches to achieve the predicted shape changing behavior in 4D printed structures [16]. In many cases, current structures are designed in a trial-and-error approach and the final shape is validated using simulation after AM [21]. Especially in FFF direct 4D printing, computational models are often used for reference and to gain some design intuition but are only rarely used for predictive purposes [22]. As structures become more complex, this trial-and-error approach often does not yield the desired results and significantly increases the effort required to design 4D printed structures. Direct 4D printing presents a significant challenge in this context since the AM pre-strain in the structure is strongly dependent on the entire manufacturing process and must be known at the start of the simulation. These circumstances create a need for modeling approaches that are capable of imparting programmed shapes at a material level to predict the shape and properties of the final structure.

One structure that has been analyzed extensively, both from a manufacturing and simulation standpoint, in the context of FFF direct 4D printing is bilayer actuators [23–26]. FFF direct 4D printed bilayer actuators can be manufactured using a wide variety of materials, including Polylactic Acid (PLA), Polyethylene Terephthalate Glycol (PETG), Acrylonitrile Butadiene Styrene (ABS), and High Impact Polystyrene [27]. Consequently, there are a variety of modeling approaches to simulate the shape change behavior of bilayers. A traditional approach is using Timoshenko bi-metal thermostats [28] to calculate the curvature and pre-strain based on bilayer benchmark samples [11, 12]. This pre-strain can be introduced into a finite element analysis (FEA) or an analytical model [12, 15, 29]. In both modeling approaches,

recent work has shown the use of laminate-based modeling to represent multimaterial structures and AM layer effects [30, 31]. In FEA, custom linear and nonlinear thermal expansion coefficients are by far the most common modeling approach [12, 30–36]. By using a linear, thermal expansion coefficient, the structure deforms continuously to its target shape as the temperature increases and would keep deforming unless the temperature increase is stopped [30, 33, 37, 38]. For nonlinear thermal expansion, this deformation can be brought much closer to an actual release of stored pre-strain in a shape memory cycle [30]. However, this does not allow the modeling of an initial or subsequent shape memory cycle [30, 37, 38], and it further complicates the simulation of time-temperature dependent mechanical properties. Specifically, in the case of the sequential activation of multiple materials, a thermal-expansion-based model only considers expansion, it does not consider the time-temperature dependent mechanical interactions between the different materials [22]. Due to the changing stiffnesses of the individual materials with temperature and the interaction of the shape memory effect and thermal expansion, a different modeling approach is required.

There are many different types of modeling approaches for the behavior of SMPs in general applications [39]. These models can be split into three main categories according to Huang *et al* [40]: (1) rheological models, (2) phase transition models, and (3) models combining viscoelasticity and phase transitions. Rheological models include time-temperature dependent viscosity and modulus parameters, the most basic example is the linear viscoelastic model. Phase transition models focus on decomposing the polymer into multiple continuous structures or phases. Generally, a frozen glassy state phase is combined with an active, rubbery state phase. The combined models use concepts from the two previous approaches in one. One of the most widely spread rheological models is the Generalized Maxwell model [40], this model is often combined with the Prony series [41, 42] and the Williams–Landel–Ferry (WLF) equation [43]. The Prony series describes the individual elements present in a generalized Maxwell model. The great advantage of using Prony's method is that a frequency signal can be transformed into a time signal via the Prony formulation. This can be leveraged to determine complex material behavior with a relatively low number of samples. Usually, the Prony series is found via relaxation tests or frequency sweeps of the desired material at different temperatures [44]. This allows accurate simulation of the shape memory cycle, but it is not directly set up to handle specific applications such as direct 4D printing, which often leads to multi-stage simulations [26].

In this paper, a thermo-viscoelastic laminate-based FE modeling approach is proposed. The approach enables the design and simulation of direct 4D printed structures manufactured using FFF. Contrary to the more typical thermal-expansion-based modeling approach, the proposed model initializes the SMP in a programmed state based on an AM pre-strain. This creates realistic shape memory behavior based on a modified generalized Maxwell model. To accurately model the decoupling of deformation magnitude and the deformation strain, the deformation magnitude is modeled in the material

model via the AM pre-strain and the deformation mode is controlled by modeling the AM layers as explicit plies in a laminate. The proposed material model and simulation approach are validated using several benchmark examples. First, three basic objects demonstrating bending and twisting as well as a direct 4D printed PLA-PETG bilayer are presented. The PLA-PETG bilayer undergoes sequential material activation on heating due to the different glass transition temperatures of its two component materials. This bilayer is a general example of a time-temperature dependent interaction between two SMPs that is not straightforward to model with the typical thermal-expansion-based approach. The last benchmark example is a direct 4D printed PLA wave spring [25] that is simulated to demonstrate the model's accuracy in predicting the final deployed shape and the mechanical properties of the wave spring after deployment for a range of design parameters. Finally, the capability of the approach to enable design space exploration is shown by conducting a parameter interaction analysis of the main design parameters of the wave spring. This demonstrates that the modeling approach provides designers with new information about design parameter trade-offs that may be non-intuitive due to the relatively new area of design for 4D Printing.

2. Methods

2.1. Material model

2.1.1. Laminate-based FE modeling of FFF direct 4D printing.

The proposed approach consists of six key steps, as shown in figure 1. In step (a), a series of samples is used to determine the curvature programmed into the AM part using direct 4D printing. In step (b), through a combination of DMA testing and curvature analysis of AM samples, the Prony series of the desired material and the pre-strains based on AM process parameters can be determined. Step (b) represents the core component of the modeling approach in the form of a modified generalized Maxwell model in combination with a WLF and Arrhenius function. In steps (c)–(e), a laminate layer modeling approach is chosen to represent the AM direction and layering present in the FFF process, these laminates are then patterned and modeled in an FE environment using 2D shell elements. Finally, the time-temperature dependent properties of the direct 4D printed structure can be analyzed in an FE simulation in step (f).

2.1.2. Generalized Maxwell model with pre-strain.

Figure 2(a) shows a typical 'hot programming' shape memory cycle. Normally, a generalized Maxwell model is initialized at zero stress and zero strain, indicating that the polymer is in its original shape. The goal is to modify a generalized Maxwell model to allow for an initialization in the programmed state as marked by the star in figure 2(a). The generalized Maxwell model in figure 2(b) consists of a single, linear Hookean element k_0 , and j Maxwell elements consisting of a linear Hookean element of stiffness k_j and a Newtonian dashpot,

or viscous element, with a relaxation time of τ_j [45]. Linear elements are chosen as the typical strain ranges for direct 4D printing are at around 5% [46]. The relaxation time is modulated using a scaling factor, α_T , based on the WLF and Arrhenius equation [47]. Thermal expansion is also considered using a thermal element with an expansion coefficient of α . The general model is described in equations (1) and (2), the incremental form is described in equations (3)–(6) and the modifications that enable the new initialization point during initialization are described in equations (7) and (8).

The governing equations of the generalized Maxwell model are modified to include an internal stress σ_0 in the elastic element k_0 in equations (1) and (2):

$$\sigma(t) = \sigma_0 + C_0 : \frac{d\epsilon}{dt} + \sum_j \sigma_j \quad (1)$$

$$\frac{d\sigma_j}{dt} + \frac{1}{\tau_j} \sigma_j = C_j : \frac{d\epsilon}{dt} \quad (2)$$

where $\sigma(t)$ is the total stress, σ_j and C_j are the internal stress and stiffness of the j^{th} Maxwell branch and C_0 is the stiffness of the elastic element k_0 . The stress of the elastic element k_0 is normally calculated based on its stiffness C_0 and the total strain $\epsilon(t)$, the modification in equation (1) allows non-zero initial stress states without removing the model's capability to model future shape memory cycles. To better illustrate how the modeling approach modifies the generalized Maxwell model in the FE simulation, the stress is calculated in its incremental form in equations (3)–(5):

$$\sigma^{n+1} = h_0^n + \sum_j A_j h_j^n + \left(1 + \sum_j M_j\right) [C_0 \Delta\epsilon] \quad (3)$$

$$A_j = \exp\left(\frac{-\Delta t}{\alpha_T \tau_j}\right) \quad (4)$$

$$M_j = \frac{k_j}{k_0} \cdot \frac{\alpha_T \tau_j \left(1 - \exp\left(\frac{-\Delta t}{\alpha_T \tau_j}\right)\right)}{\Delta t} \quad (5)$$

where σ^{n+1} is the stress of the time step $n+1$ in the FE simulation based on the residual stress of the linear Hookean element h_0^n and the residual stresses of the Maxwell elements h_j^n from the previous time step n . The decay A_j (equation (4)) of the residual stresses h_j^n in the Maxwell elements due to the damper is based on the relaxation time τ_j , the current time step Δt , and the relaxation time scaling factor introduced by the WLF and the Arrhenius functions α_T . The newly introduced stress of the current time step is based on the stiffness C_0 of the linear Hookean element k_0 and the strain $\Delta\epsilon$ of the element in the current time step. The individual Maxwell elements are modeled based on a normalization factor M_j (equation (5)) used to adapt the stiffness matrix C_0 of the elastic element k_0 to the individual Maxwell elements. Finally, equation (6):

$$\log_{10} \alpha_T = \begin{cases} C_3 \left(\frac{1}{T} - \frac{1}{T_g}\right) & \text{if } T < T_g \quad \text{Arrhenius} \\ \frac{-C_1(T-T_g)}{C_2+(T-T_g)} & \text{if } T \geq T_g \quad \text{WLF} \end{cases} \quad (6)$$

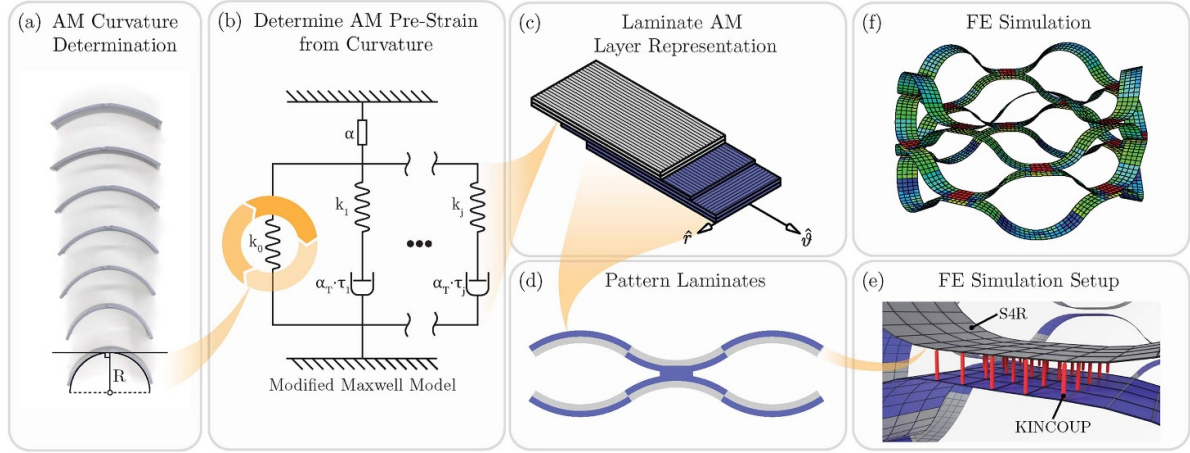


Figure 1. (a) Determination of the AM pre-strain using bilayer samples with varying AM parameters. (b) Relate curvature to an AM pre-strain implemented in a modified Maxwell model. (c) AM layers and layer direction representation using a laminate modeling approach. (d) Laminates are patterned to create complex geometry. (e) The pattern is represented by shell elements in a FE simulation environment. (f) Direct 4D printed structures are analyzed using FEM.

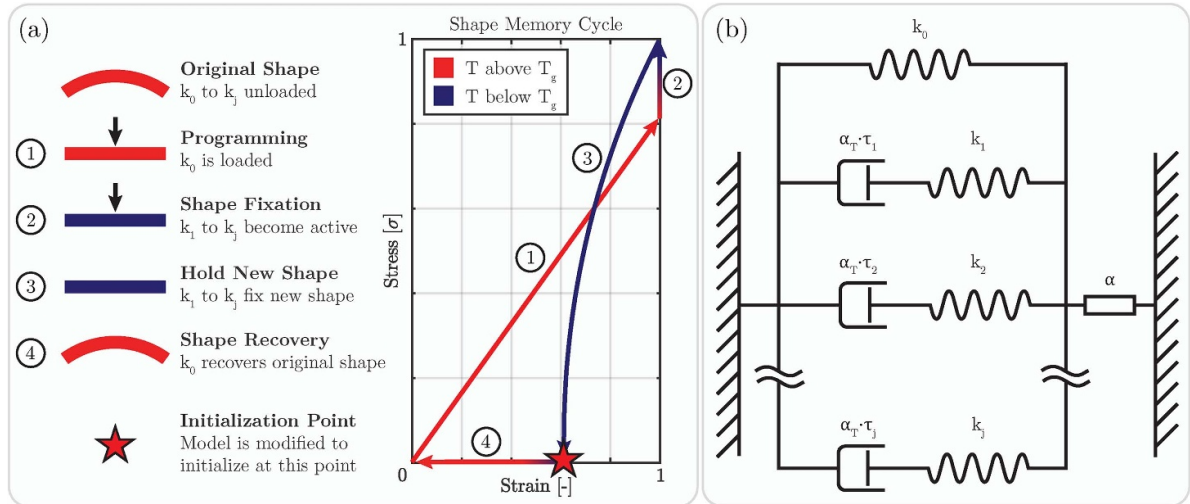


Figure 2. (a) Shape memory cycle with the active elements of the generalized Maxwell model highlighted for each phase. The new initialization point for the model is marked with a star. (b) Illustration of the generalized Maxwell model used to represent the material behavior of direct 4D printed polymers. The model consists of a single, linear Hookean element k_0 and j Maxwell elements consisting of a linear Hookean element k_j and a Newtonian damper with a relaxation time of τ_j . The relaxation times are modulated according to the WLF and Arrhenius function using the factor α_T .

defines the relaxation time scaling factor, α_T , for the relaxation times, τ_j , based on the WLF [43, 48] and Arrhenius [49] equations. If the current temperature T is higher than the glass transition temperature T_g , the WLF equation is applied. For temperatures below the glass transition temperature, the Arrhenius equation is applied. The scaling is given by the constants C_1 and C_2 for the WLF equation and the constant C_3 for the Arrhenius equation.

To allow for the introduction of an AM pre-strain and a model initialization in a programmed state, as marked by the star in figures 2(a), (a) residual stress term h_0^n is introduced for the purely elastic element k_0 as shown in equation (1). As illustrated by figure 2(a), the element k_0 is responsible for shape recovery in the shape memory cycle. A non-zero initial stress

state in the element k_0 introduces the capability to initialize the polymer in a programmed state. As shown in equation (7):

$$h_0^n = C_0 \varepsilon_0 \quad (7)$$

this initial stress state at time $t = 0$ is determined based on the AM pre-strain ε_0 . In subsequent steps it is stored in a state variable, like all the other residual stresses in the system. If the initial stress state is only introduced in h_0^n , the FE solver would have to solve for a non-zero global stress state in the first step of the simulation. To prevent this, the residual stresses in all Maxwell elements are balanced according to the initial stress in the purely elastic element k_0 , as shown in equation (8):

$$\mathbf{h}_j^n = -\mathbf{h}_0^n \cdot \frac{k_j}{\sum_j k_j} \quad (8)$$

This balancing step enforces zero resulting stress in the first step of the simulation as the Maxwell elements balance out the stored pre-strain in the elastic element k_0 . As shown in equation (9):

$$\mathbf{C}_0 = \frac{k_0}{1-\nu^2} \cdot \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & 1-\nu \end{bmatrix} \quad (9)$$

a plane stress condition is introduced in the stiffness matrix to allow for the simulation of 2D shell elements. The stiffness matrix for the purely elastic element is defined via the tensile stiffness k_0 and Poisson's ratio ν . The described model is implemented in a custom Abaqus UMAT subroutine using Fortran 77 [50]. The required inputs for the model are described in the supplementary information and in the UMAT file itself.

2.1.3. Simulation parameters. All simulations are conducted in Abaqus/Standard using four-node quadrilateral shell elements (S4R) with reduced integration. The laminate layups are modeled explicitly and identical adjacent layers are grouped into thicker single layers. The explicit laminate definition is added via the 'COMPOSITE' option within a regular '*Shell Section'. Laminate layers are assigned a material, given a thickness based on the AM layer thickness and are oriented based on the AM fabrication direction relative to the local material direction. Perfect bonding between individual materials and layers is assumed. To simplify the orientation, the element material orientation is aligned with the direction of maximum AM pre-strain. Materials are initialized in Abaqus/Standard as a '*Material' with the option '*User Material' and the '*Depvar' option set to create six state variables for the linear spring element k_0 and each spring element k_j in the Maxwell branches. Additionally, a density is added via '*Density' and thermal expansion is included into the material behavior with the '**Expansion' option. Due to the time-dependent behavior being handled in the UMAT subroutine, '*Static' load steps can be used to simulate the viscoelastic behavior. Geometric nonlinearity is considered through the activation of the 'NLGEOM' option. The convergence criteria for all steps are maintained at the Abaqus default settings. Specific element and time step parameters are discussed alongside each specific benchmark example in the following sections.

2.2. Material characterization

2.2.1. Prony series characterization. The thermo-viscoelastic response of the polymer is characterized by using the well-established time-temperature superposition principle [43, 51]. For this work two polymers are characterized, Prusament PLA [52] and Extrudr PETG [53]. At least three ASTM-D638 Type IV test specimens are fabricated per material. The specimens are then subjected to frequency

sweeps in the range of 1 – 18 Hz. The frequency sweeps are conducted at 5 °C intervals between 23 °C and 85 °C for the PLA specimens and 23 °C and 100 °C for the PETG specimens. The temperature ranges are set to end around 20 °C above the expected glass transition temperature T_g of the individual polymers. This ensures accurate measurement of the long-term Young's modulus k_0 , which characterizes the polymers' behavior in the rubbery phase. All tests are conducted on an Instron ElectroPuls E3000 using a 5 kN Dynacell load cell, an Instron environmental chamber, and Instron fatigued mechanical wedge action tensile grips. During testing, Instron WaveMatrix is used to calculate the storage modulus E' , the loss modulus E'' and the phase shift $\tan \delta$ between the two moduli for each test specimen at different frequencies and temperatures.

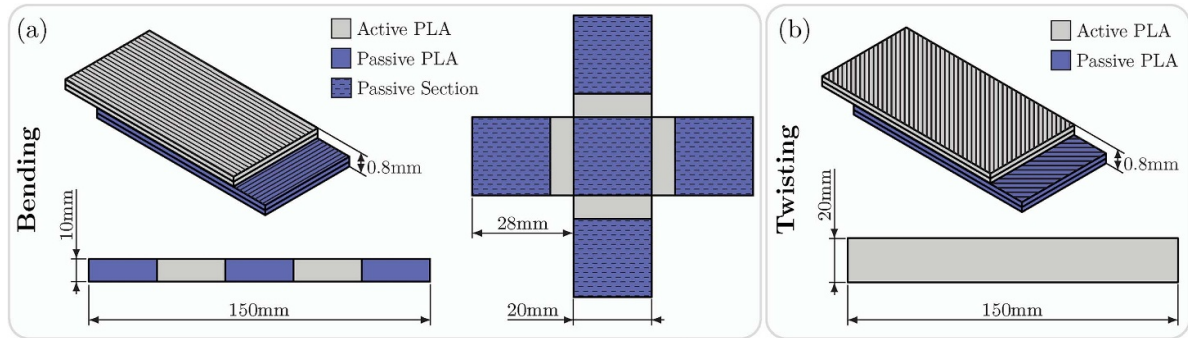
DMA data is recorded for each frequency at each temperature. The glass transition temperature T_g is determined based on the observed peak in the phase shift $\tan \delta$ [54]. The frequency dependent storage modulus is subsequently shifted according to the WLF function above T_g and according to the Arrhenius equation below T_g ; see equation (6) and section A.1 in the SI for more information. This allows for better fine tuning of the shift and a more accurate fit of the relevant parameters. This also simplifies the determination of the constants C_1 and C_2 . As shown by Ferry [43], if the reference temperature T_0 is equal to the glass transition temperature T_g , the constant C_1 is equal to 17.4 and C_2 is equal to 51.6. Based on three test specimens per material, an average master curve is derived, which describes the relationship between frequency and storage modulus. Finally, the Prony series is fitted to the master curve in the frequency domain using the least squares method to obtain the final material parameters (see section A.1 in the SI).

2.2.2. AM pre-strain determination. The AM pre-strain is determined using several bilayer benchmark samples, which is a common approach in literature [11, 12, 31, 33]. The bilayer samples consist of seven layers each, which includes four passive layers and three active layers for both PLA and PETG. Six samples are fabricated for each nozzle and activation temperature. The samples are 25 mm long and 5 mm wide and are fabricated with a layer height of 0.1 mm on an E3D Motion System and ToolChanger (E3D, Chalgrove, United Kingdom) using E3D Hemera extruders. For this work, samples with AM process parameters according to table 1 are evaluated. After fabrication, the samples are deployed at 75 °C for PLA and 120 °C for PETG. The PLA is activated in a hot water bath, a uniform temperature distribution in the bath is achieved through circulation, this circulation is stopped immediately prior to the activation to prevent the current influencing the deployment result. The PETG is activated in a heated chamber, the chamber contains a fan to ensure an even temperature distribution. The radius, and thus curvature, of the activated bilayers is then determined based on high resolution images of the samples containing a reference scale.

The AM pre-strain is determined both by using the theory behind Timoshenko bi-metal thermostats [28] and by fitting a

Table 1. AM and activation parameters used during the fabrication of the benchmark bilayer samples.

Parameter		PLA	PETG
h_{Layer}	[mm]	0.1	0.1
T_{Nozzle}	[°C]	{195, 200, 215}	{205, 210, 215, 220}
$f_{\text{Volumetric}}$	[mm ³ s ⁻¹]	15	17
Fan Speed	[%]	100	0
$T_{\text{Activation}}$	[°C]	{75, 90}	120
Nozzle diameter	[mm]	0.4	0.4
T_{Ambient}	[°C]	22	22
T_{Bed}	[°C]	60	70

**Figure 3.** (a) AM laminate layup of the two bending objects, active layers are oriented to induce maximum bending. (b) AM laminate layup of the twisting tube design. The active layer is oriented at 45 degrees relative to the long edge to ensure maximum twisting.

simulated bilayer to the target curvature using a pattern search algorithm to enable benchmarking of a basic single material bilayer. In both cases, it is assumed that the passive layers do not contain any pre-strain at the start of the deployment. The simulated bilayers are fixed at one end and then deployed by raising their nodal temperature to the specified activation temperature $T_{\text{Activation}}$. The optimization models the pre-strain in the active material of the bilayer as a variable and seeks to minimize the absolute difference in curvature between the AM and simulated sample to find the pre-strain that best maps to the desired curvature. Additional details on this fitting procedure are provided in the SI in section A.2.

2.3. Material model validation

2.3.1. Basic shape change and time-temperature dependent behavior. The basic functionality of the model is shown using three sample objects. The objects are chosen such that they illustrate the ability of the model to simulate active and passive sections as well as bending and twisting behavior in direct 4D printed bilayers. All bilayers consist of four active and four passive 0.1 mm layers of PLA. The specific layup and AM fabrication directions of the individual objects are shown in figure 3. Two objects use bilayer bending to create a box and a braid (figure 3(a)). The third object uses bilayer twisting to create a tube from an initially flat configuration (figure 3(b)).

To validate the simulation of the time-temperature dependent material behavior, a direct 4D printed sequentially activating multimaterial bilayer consisting of both PLA and PETG

is designed. The bilayer consists of three active 0.1 mm layers of PLA and five active 0.1 mm layers of PETG. The PLA is fabricated at a nozzle temperature of 200 °C and the PETG is fabricated at 205 °C. The remaining process parameters are identical to the ones in table 1. The specimen is rectangular with a length of 50mm and a width of 10mm. Both the PLA and the PETG are programmed during AM through process parameter manipulation. The AM direction is aligned with the long edge of the bilayer for all layers. Upon heating, the PLA layers activate first at $T_g = 65$ °C. The PETG layers activate later when their higher glass transition temperature of $T_g = 85$ °C is reached. The bilayer is initially heated to a 50 °C starting temperature and it is then heated to 95 °C at a rate of 4 °C min⁻¹ to enable controlled sequential activation.

The three sample objects and the sequential bilayer are simulated in Abaqus/Standard using four-node quadrilateral shell elements (S4R) and step sizing according to table 2. The models use the material parameters shown in tables B.1 and B.2 in the supplementary information. The respective AM pre-strain is based on the analysis from section 2.2.2 and the corresponding results in section 3.1.2. The sequential bilayer and tube are constrained along one of their short edges using a fully fixed (*encastre*) boundary condition. The braid is constrained at the center of the middle bilayer using the same fully fixed boundary condition. The center square section of the box is pinned to stop unwanted motion. The Abaqus input files for all four simulations can be found in the supplementary information along with the Fortran subroutine used to model the material; see section C.1 in the SI.

Table 2. Element characteristics and simulation parameters for the three sample objects and the sequential bilayer.

Parameter	Box	Braid	Tube	Sequential Bilayer
Element count	2640 S4R	1500 S4R	3000 S4R	125 S4R
Characteristic element length	1 mm	1 mm	1 mm	2 mm
Analysis type	Static, general	Static, general	Static, general	Static, general
NLGEOM	Enabled	Enabled	Enabled	Enabled
Time period	30 s	30 s	30 s	750 s
Min step size	1×10^{-4} s	1×10^{-15} s	1×10^{-4} s	1×10^{-4} s
Max step size	1 s	1 s	1 s	20 s

Table 3. Element characteristics and step sizes for the direct 4D printed wave spring.

Parameter	Heating	Steady State	Cooling	Loading
Element count	2880 S4R / 375 KINCOUP			
Characteristic element length	1.13 mm			
Analysis type	Static, general	Static, general	Static, general	Static, general
NLGEOM	Enabled	Enabled	Enabled	Enabled
Time period	30 s	10 s	10 s	10 s
Min step size	1×10^{-4} s	1×10^{-4} s	1×10^{-4} s	1×10^{-4} s
Max step size	1 s	1 s	1 s	2 s

2.3.2. Direct 4D printed wave spring. To further validate the proposed model and show how it can be applied in design, it is used to simulate the deployment of a direct 4D printed wave spring proposed in previous work [25]. The wave spring uses direct 4D printing and a bilayer design to enable a flat configuration for fabrication and subsequent deployment. The wave spring is simulated in Abaqus/Standard using four-node quadrilateral shell elements (S4R) in the individual wave rings and kinematic couplings (KINCOUP) in the connectors between the wave rings (see figure 4(e) and table 3). The mesh and input file for Abaqus are generated using a custom MATLAB script to model the wave spring based on the five initial input parameters shown in figures 4(a)–(d). This approach is chosen over automatic mesh generation strategies as it enables a more controlled mesh generation and geometry-specific model abstractions, facilitating a fully parametric FE simulation. These abstractions also result in a simulation model that is less computationally expensive compared to other methods, such as full solid modeling. The input file for each spring, characterized by the five unique parameters, is generated, submitted, and the resulting Abaqus outputs are evaluated using MATLAB to facilitate the subsequent design parameter study.

The number of AM layers in the bilayer and thus the thickness of the bilayer is given by the parameter n_{Bilayer} . An individual wave ring within a wave spring consists of multiple bilayers and the parameter n_{Waves} defines the number of waves per wave spring. A wave spring consists of multiple wave rings that are linked by a passive connector. The number of wave rings in the wave spring is defined by the parameter n_{Layers} . The outer dimensions of the wave spring are defined by the outer diameter d_{Outer} and the wave ring width w . The

AM layer height is fixed at 0.1 mm. The individual AM layers are defined according to the convention shown in figure 4(a), where laminate layers are modeled as the shell elements. The active layers and the passive carrier layer are oriented along the $\hat{r}$ axis whereas the passive layers are oriented along the $\hat{\vartheta}$ axis. To ensure consistent layer orientation, the material orientation for all shell elements is aligned with the tangential direction $\hat{\vartheta}$.

The simulation is run in cylindrical coordinates. The constrained nodes at the bottom of the wave spring are constrained in the tangential direction $\hat{\vartheta}$ and in the z -direction. The simulation is divided into four steps. In the first step, the wave spring is heated from an initial nodal temperature of 75 °C release the pre-strain in the elements and simulate the deployment. The wave spring is then kept in a steady state at 75 °C. The wave spring is subsequently cooled back to 22 °C in the third step. In the final step, the wave spring is loaded to determine its mechanical properties. The loading at the top is introduced as a displacement in the negative z -direction in this step; no additional constraints are added to the system. An example input file for the simulation in Abaqus can be found in the supplementary information along with the Fortran subroutine used to model the material; see Section C.2 in the SI.

The simulated wave spring is subsequently compared to its AM counterpart. Five physical wave springs are fabricated and tested according to the method presented in previous work [25] using the design parameters stated in table 4. Both the mean deployed free height and mean force response to loading are compared. To further validate the general behavior of the model and show a possible application, a series of 720 FE simulations is conducted to analyze the interactions of the main

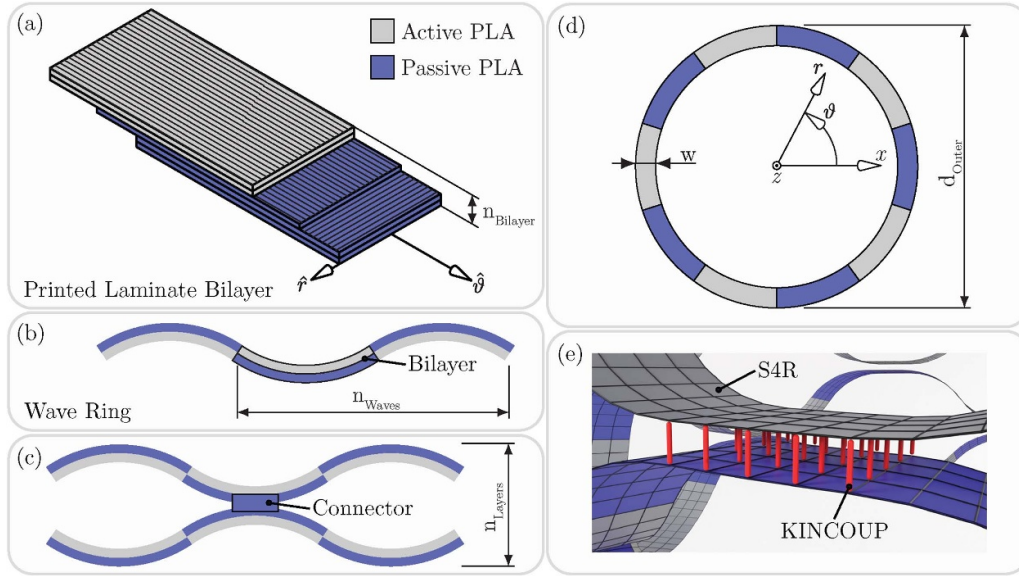


Figure 4. (a) AM laminate layup for the simulation in a cylindrical coordinate system. (b) The parameter n_{Waves} defines the number of waves per wave ring. (c) The number of wave rings in the wave spring is defined by the parameter n_{Layers} . (d) The outer dimensions of the wave spring are defined by the outer diameter and the wave ring width w . (e) The FE mesh for the simulation consists of four-node quadrilateral shell elements and kinematic couplings.

Table 4. Design parameters of the direct 4D printed wave spring used for comparison with the simulation model and the set of parameters used for the interaction analysis of the main design parameters.

Parameter		Physical Prototype	Interaction Analysis
d_{Outer}	[mm]	70	{38, 54, 70, 86}
w	[mm]	5	{2, 5, 8, 11}
n_{Bilayer}	[—]	3	{1, 2, 3, 4, 5}
n_{Waves}	[—]	5	{3, 4, 5}
n_{Layers}	[—]	4	{2, 3, 4}

design parameters in the direct 4D printed wave spring. Based on a series of values for each design parameter (see table 4), the resulting spring stiffness of each simulated wave spring is used to analyze the statistical relation between the design and solution space considering main effects and first order interaction effects. The response for a specific combination of parameters is an averaged value based on every version of the wave spring that shares the two design variables being compared, which in this case is between 36 and 80 wave springs. Five design parameters are compared in the context of this work. The analysis is conducted using the MATLAB function *interactionplot*.

3. Results

3.1. Material characterization

3.1.1. Prony series characterization. The full long-term mechanical properties of both PLA and PETG are shown in table B.1 in the SI. All presented properties of both PLA and PETG are determined based on the methodology stated in section 2.2. The glass transition temperature T_g is determined

based on the $\tan \delta$ peak shown in figure 5. PETG has a rubbery phase modulus k_0 of 8.093 MPa and PLA has a rubbery phase modulus k_0 of 10.598 MPa according to the measured data. A total of 23 Maxwell elements are chosen for both PLA and PETG to fit the Prony series. The experimentally determined storage modulus $E'(t)$ and the corresponding results of the Prony series fit are shown for both materials in figure 5. Both materials show good agreement between the measured material properties and the fitted material model. The resulting elastic element stiffness, k_i , and the damping relaxation time τ_i of the individual Maxwell elements are shown in table B.2 in the SI. A sensitivity analysis of the model accuracy versus the number of Maxwell elements can be found in figure B.1.

3.1.2. AM pre-strain characterization. Table 5 shows the resulting AM pre-strain from the samples tested according to the method described in section 2.2.2. Both PLA and PETG show a pre-strain that correlates with the nozzle temperature. Lower AM temperatures show larger pre-strain than higher temperatures in both materials. The Timoshenko bi-metal thermostats approach shows slightly lower values than what is found by pattern search. The maximum deviation between the two approaches amounts to 0.231% for PLA and 0.938% for PETG.

3.2. Material model validation

3.2.1. Basic shape change and time-temperature dependent behavior. The initial fabricated and activated states as well as the simulated activated state of all three sample objects are shown in figure 6. The simulated activated state in figure 6(a) shows good agreement with fabricated sample. It is able to replicate the two unique curvatures, the touch condition at

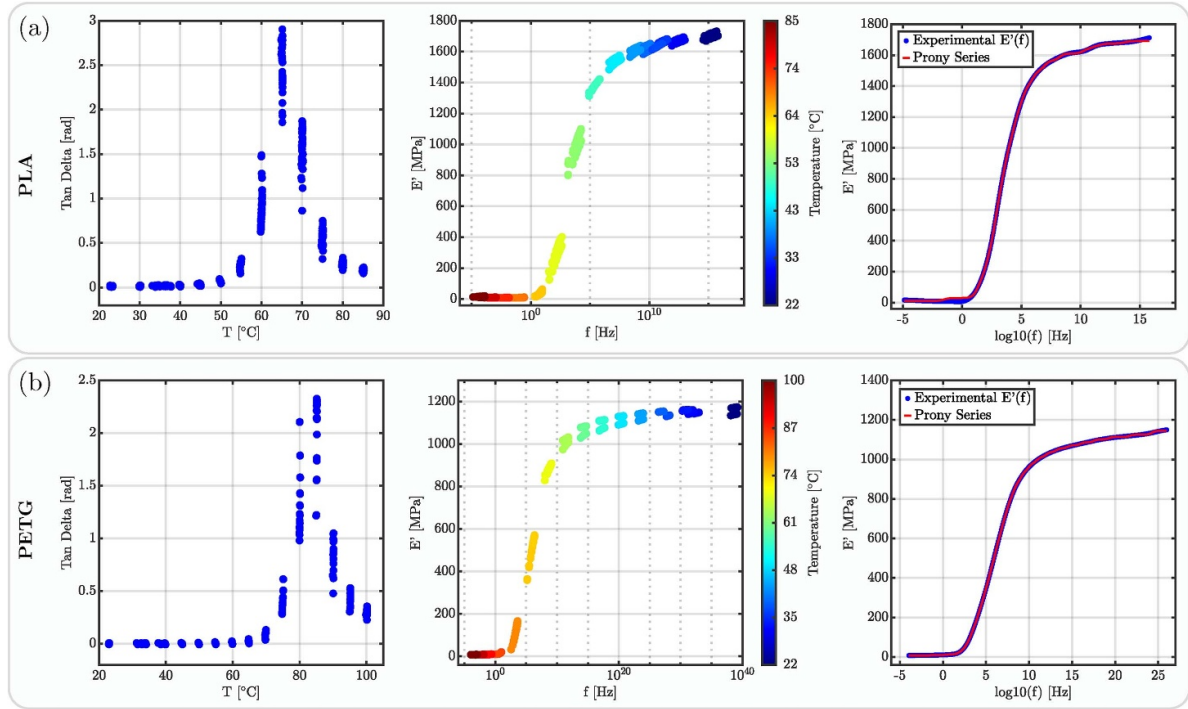


Figure 5. Measured phase shift $\tan \delta$, individual experimental storage modulus E' measurements from frequency sweeping and final master curve with Prony series of both PLA (a) and PETG (b). The frequency sweeps are colored based on the measurement temperature in Celsius.

Table 5. Resulting AM pre-strain ε_{11} based on the AM parameters from table 1. The pre-strain is shown for the pattern search and Timoshenko bi-metal thermostats approach presented in section 2.2.2.

Material	T_{Nozzle} [°C]	$T_{\text{Activation}}$ [°C]	κ [–]	ε_{11} Pattern Search	ε_{11} Timoshenko	Difference[%]
PLA	195	75	0.11766	0.05618	0.05605	0.232
PLA	200	75	0.08815	0.04206	0.04199	0.167
PLA	200	90	0.18088	0.10350	0.10290	0.583
PLA	215	90	0.13933	0.07957	0.07926	0.391
PETG	205	120	0.07	0.06029	0.05973	0.938
PETG	210	120	0.0350	0.03014	0.02987	0.904
PETG	215	120	0.0215	0.01851	0.01835	0.872
PETG	220	120	0.0165	0.01421	0.01408	0.923

the center of the sample and the large deformation at either end creating an almost fully closed loop. The simulated active state of the second bending sample in figure 6(b) also shows good agreement with the fabricated sample where all passive sections remain undeformed and the bending is localized in the designed hinges. The simulated activated state of the twisting sample in figure 6(c) does not agree fully with the activated state of the fabricated sample. The simulated sample does not twist as tightly as the AM fabricated sample and it reaches its final state at around 2.5 axial rotations instead of the close to three full axial rotations of the AM fabricated sample.

Results for the deployment of both the AM fabricated and simulated sequential PETG-PLA bilayer are shown in figure 7 and table 6. In the AM fabricated bilayer, the PLA activates first as its glass transition temperature $T_g = 65^\circ\text{C}$ is passed (figure 7(b)). Initially the bilayer displaces slowly up to around the 350 s mark, the displacement accelerates and reaches a maximum at $t = 410$ s before the T_g of PETG ($T_g = 85^\circ\text{C}$) is

reached. The bilayer then reverses direction and flips to the other side as the sample is further heated towards 100°C . The maximum average angular displacement towards the PLA is $-47.8^\circ \pm 9.3^\circ$ and towards the PETG the maximum average displacement is $41.0^\circ \pm 38.8^\circ$ (figures 7(a) and (b)). The simulated bilayer also activates when the glass transition temperature of PLA is reached. It also shows the characteristic initial small displacement and subsequent increasingly larger displacement from $t = 350$ s. The maximum displacement of -58.2° is reached at $t = 434$ s. The simulated bilayer then also reverses direction and comes to rest at a final displacement at 39.6° after $t = 570$ s. The root mean square error (RMSE) between $t = 0$ s and $t = 450$ s is equal to 3.6° . The RMSE between $t = 450$ s and $t = 750$ s is equal to 36.2° . The RMSE for the full time span is equal to 21.6° . A video comparing the activation of the AM fabricated and simulated bilayer can be found in the supplementary information (video S3).

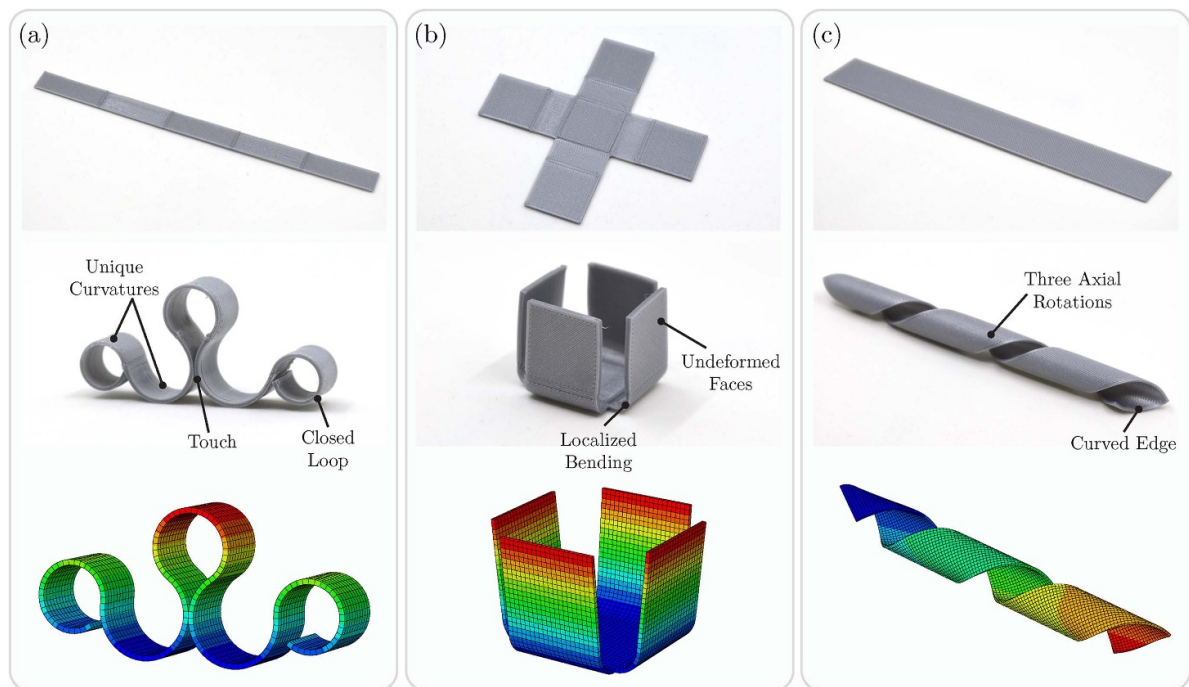


Figure 6. Fabricated initial and activated states as well as the simulated activated state of all three sample objects. The first object contains a touch condition, unique curvatures and fully closed loop. The second object exhibits passive and active sections with strong localized bending. The third object twists around its own axis three times in the experiment, whereas the simulated sample twists 2.5 times.

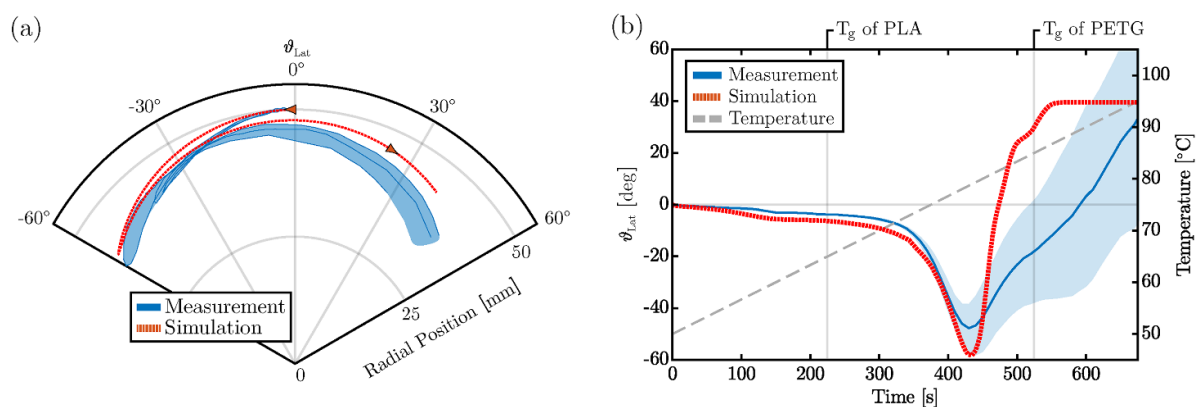


Figure 7. (a) Comparison of the path traced by the end point of the bilayer upon activation for both the AM fabricated and simulated sequential bilayer. (b) Time-temperature dependent angular displacement of the fabricated and simulated sequential bilayer. The mean and standard deviation of three individually measured bilayers is shown.

Table 6. Comparison of activation characteristics of the AM fabricated and simulated sequential PETG-PLA bilayer.

Result		Experimental	Simulated
Displacement PLA	[deg]	-47.8°	-58.2°
Std. Dev.	[deg]	9.3°	—
Time	[s]	410	434
Displacement PETG	[deg]	41.0°	39.6°
Std. Dev.	[deg]	38.8°	—
Time	[s]	700	570
RMSE ($0 \leq t \leq 450$ s)	[deg]		3.6°
RMSE ($450 \leq t \leq 750$ s)	[deg]		36.2°
RMSE ($0 \leq t \leq 750$ s)	[deg]		21.6°

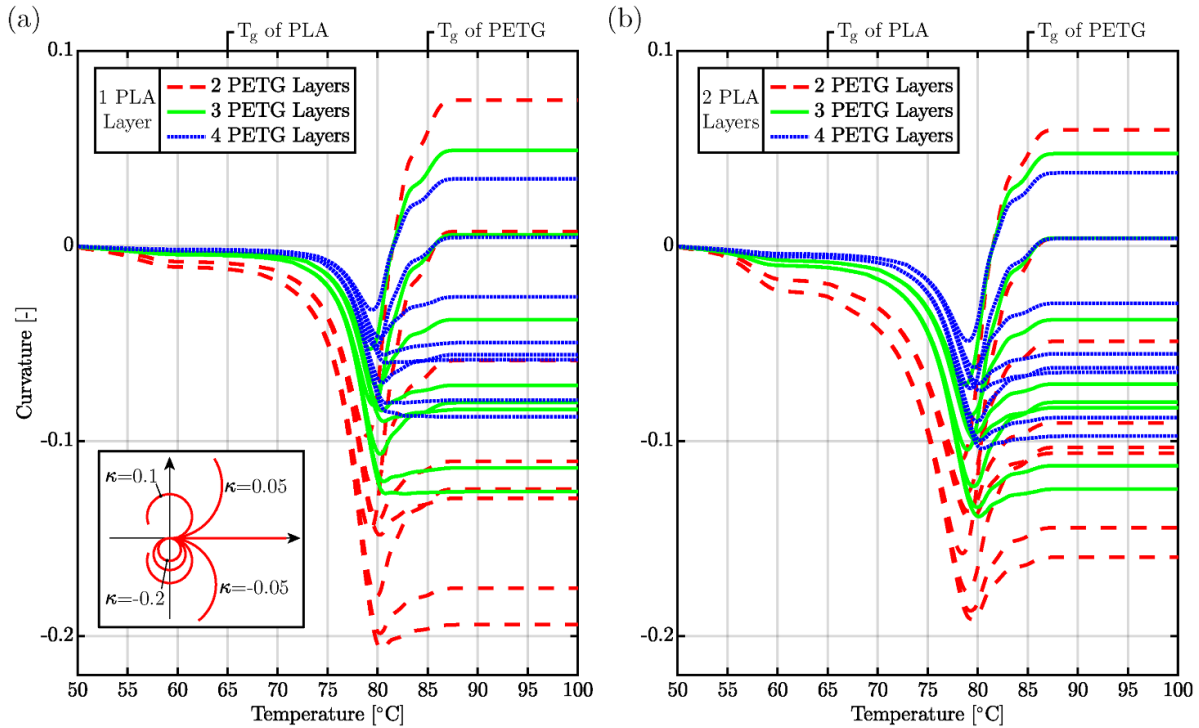


Figure 8. Curvature range of the simulated PLA-PETG bilayer samples as a function of temperature in Celsius. A negative curvature indicates that the sample is curling towards the active PLA layer as shown in (a), a positive curvature indicates that the sample is curling towards the active PETG layer. One active PLA layer is combined with different AM pre-strains and PETG layer counts in (a) and likewise in (b) for two active PLA layers.

Additionally, the design space is explored by using the same bilayer dimensions of 50 mm by 10 mm as described in section 2.3.1 and combining it with the six individual pre-strains shown in table 5 in a variety of layups. The resulting temperature-dependent curvature is shown in figure 8. In figure 8(a) all combinations for one active PLA layer are shown and in figure 8(b) all combinations for two active PLA layers are shown. All bilayers initially curl towards the active PLA layers and show at least some reversal of this initial curvature when the PETG is activated. Most of the bilayers do not flip back past their initial position and curve in the other direction. The changing pre-strains and layer counts influence the behavior of the bilayer in a variety of ways. Changing the number of PETG layers and pre-strain in both figures 8(a) and (b) modulates the range of motion as well as the activation times.

3.2.2. Direct 4D printed wave spring. Figure 9 shows the results of the deployment and static characterization comparison conducted according to section 2.3.1. The wave spring is simulated using the pre-strain values resulting from the pattern search, as listed in table 5. The simulated wave spring (figure 9(b)) shows good agreement with the general shape of the AM fabricated wave spring shown in figure 9(a). As shown in previous work [25], the multiple turn wave spring deploys to 31.9 mm with a standard deviation of 2.29 mm. The simulated

wave spring shown in figure 9(b) reaches a deployed height of 32.82 mm. As shown in figure 9(e), the simulated spring shows correct shape memory behavior and AM pre-strain. Upon initial heating, the wave spring does not deploy. When the glass transition temperature of PLA is reached at $T_g = 65^\circ\text{C}$, the wave spring begins to deploy and reaches its full free height within 9 s of simulation time. The wave spring is then cooled back to 22°C without loss of the deployed shape. The wave spring can then be tested directly to determine the mechanical properties, a direct comparison of the resulting axial reaction force is shown in figure 9(f). The direct comparison of the mean reaction force of the five AM fabricated and the reaction force of the simulated wave spring at 40% axial strain shows a maximum mean axial reaction force of 43.69 N for the AM wave spring and 40.81 N for the simulated wave spring. The RMSE of the reaction force is equal to 2.3 N. Two videos showing side by side comparisons of the simulated results and AM wave spring are available in the supplementary information (videos S1 and S2).

To capitalize on the simulation for design space exploration, the interactions between individual design parameters of the fabricated state can be analyzed for a variety of wave spring configurations in the activated state. The design parameter interaction analysis shown in figure 10 is conducted according to section 2.3.1. For example, consider the interaction between the outer spring diameter and the ring width. To analyze how changing the ring width at different diameters

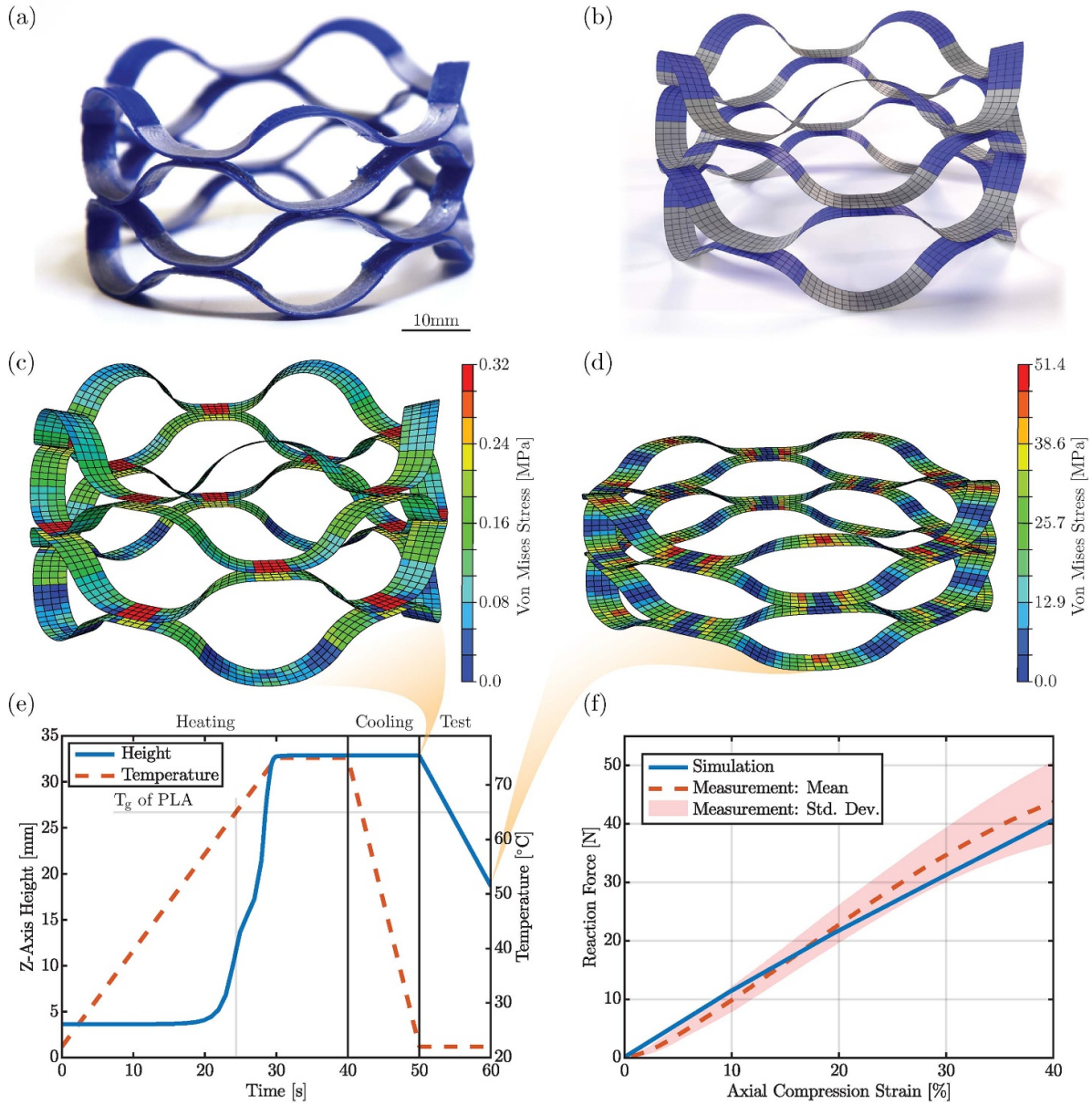


Figure 9. (a,b) Image of the deployed manufactured and simulated wave spring. (c), (d) Von Mises stress distribution in the wave spring after cooling and after loading. (e) Z-axis height in mm versus activation time in seconds and temperature in Celsius of the simulated wave spring. (f) A comparison of the reaction force versus axial compression strain curves of both the manufactured wave spring and the simulated wave spring. The mean and standard deviation of five individually measured wave springs is shown.

changes the spring stiffness, the first plot in column two, row one can be examined. The same can be done for changes in the outer diameter in the first plot in the second row. Based on this analysis, the five main design parameters show varying effects on the average final stiffness of the wave spring. The largest influence is observed when changing the number of AM layers n_{Bilayer} in the third column of figure 10, the average stiffness increases significantly when increasing the number of layers. This effect can be observed in all four plots in the third column, e.g. in the second plot increasing the number of layers from one to five causes the stiffness to increase by a factor of 7.35 for a width of 2 mm.

4. Discussion

4.1. Material characterization

The fitted Prony series for PLA and PETG (figure 5) shows good agreement with experimentally determined storage modulus E' . The method for material characterization described in section 2.2.1 worked well for both materials. The presented Prony series from table B.1 along with the long term properties and thermal properties in table B.2 enable the accurate simulation of the time-temperature dependent material behavior of both polymers. The expansion to direct 4D printing involves the inclusion of AM pre-strain. The determination of

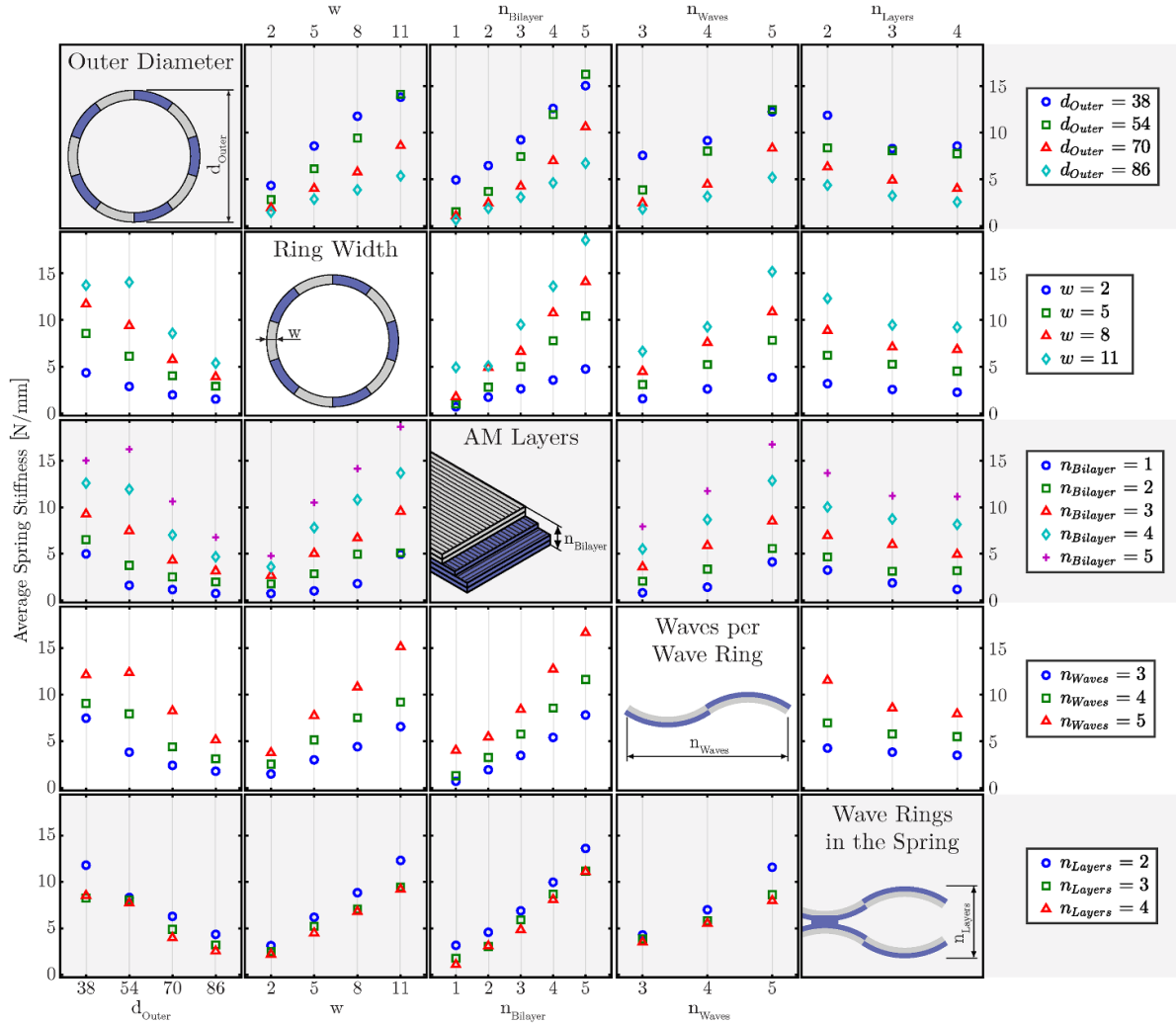


Figure 10. Interaction plot comparing the five main design parameters of the direct 4D printed wave spring to the simulated average spring stiffness in N mm^{-1} based on the described modeling approach.

the AM pre-strain through the curvature of the bilayer benchmark samples is consistent across both proposed methods. With the maximum deviation amounting to just 0.231% for PLA and 0.938% for PETG, these results provide an initial validation of the proposed model and its capability to represent AM pre-strain accurately for thin structures. The Timoshenko bi-metal thermostats approach is well known in literature [11, 12, 31, 33] as a common approach to determining AM pre-strain in direct 4D printing and thus validates the capability of the proposed model to accurately model pre-strain at a material level.

4.2. Material model validation

4.2.1. Basic shape change and time-temperature dependent behavior. The three sample objects show that the current modeling approach is capable of accurately representing large deformation bending and twisting behavior in direct 4D printed bilayers for a variety of structures. It is further capable of modeling active and passive sections. The largest deviation

from the AM shape can be observed for twisting behavior. It is likely that the simulation boundary conditions play a critical role in this discrepancy. The manufactured bilayer is activated in a hot water bath and as the sample is floating it can deform along any edge without constraints. In the simulation, this free-floating behavior is not desirable and the sample needs to be constrained on one edge to prevent random rotation and convergence issues. This constraint forces one short edge of the rectangular bilayer to remain straight, thus stopping some of the pre-strain from being released.

The analysis of the sequential bilayer in figure 7 allows for a direct comparison of the time-temperature dependent behavior of the AM fabricated sequential bilayer and the simulated bilayer. In the initial stages, the simulation accurately reflects the time-temperature dependent behavior of the bilayer as shown by the RMSE of 3.6° between 0 and 450 s. Both the fabricated (-4.13°) and simulated (-6.30°) bilayer show only small deformations around the glass transition temperature of PLA. This is mainly due to PETG still being far below its T_g , thus preventing the PLA, currently exhibiting a modulus of 12 MPa, from releasing its pre-strain, due to the significantly

higher modulus of 980 MPa that PETG exhibits at this stage. As the sample slowly approaches the glass transition temperature of PETG, the PETG begins to soften. The PLA is then able to further increase the displacement of the bilayer, creating the stepped displacement curve visible in figure 7(b). The maximum angular displacement of the simulated bilayer (-58.2°) lies at the boundary of the standard deviation of the average of the maximum angular displacement ($-47.8^\circ \pm 9.3^\circ$) of the AM bilayer on the initial activation of the PLA layers. This maximum displacement is reached at 77.3°C in the fabricated bilayer, the simulated bilayer reaches the maximum displacement at 78.9°C . The AM bilayer reaches its maximum displacement less than 2°C earlier than the simulated bilayer; the PETG's modulus has dropped to around 125 MPa by this temperature.

The main deviations between the simulation and AM fabricated sequential bilayer are observed in the final stages of the activation, as exhibited by figure 7 and the RMSE of 36.2° after the initial activation concludes at 450 s. During the first activation, the standard deviation ranges between 0° initially and 9.3° at its maximum displacement. The second activation shows larger deviations in both the final angular displacement and the temperature dependent behavior; it starts at the 9.3° from the previous step and reaches a maximum of 38.8° . While the angular displacements of 39.6° for the simulated bilayer and 41.0° for the fabricated bilayer are only 3.4% apart at 95°C , the fabricated bilayer continues to deform as the sample is heated further. This effect is not observed in the simulated sample that releases its pre-strain and then stops deforming. The final state of the simulated sample is strongly informed by the long-term rubbery modulus k_0 of both materials. If the temperature is raised far enough, all Maxwell elements converge to very short relaxation times due to the WLF function. The only remaining active element for both materials is the single linear Hookean element k_0 that makes the final state a purely elastic balance of the initial pre-strains and the respective moduli. This is one drawback of the simpler approach to the modeling of the rubbery phase, but it also decreases model complexity and computational expense. Direct 4D printing is commonly used to deploy a structure and subsequently cool it back down to fix the intended permanent shape. In this case, a linear-elastic rubbery phase (k_0) does not play a significant role in the final mechanical characterization of the structure below T_g .

Although the interlayer bonding of PLA and PETG can be up to 80% weaker than the bonding of the individual homogenous materials [55], the AM sequential bilayer did not exhibit any observable delamination between the two constituent materials. This is likely due to the low stress levels observed in the bilayer as shown in figure B.3 in the supplementary information and the slow heating regime employed to activate the bilayer.

The sequential bilayer design space exploration shown in figure 8 illustrates the capability of the approach to rapidly explore a variety of different design configurations and predict their time-temperature dependent response. Depending on the

time resolution required, the individual simulation runs for between 120 and 200 s on a mobile processor (Intel Core i7-10750H). This is assuming that a slow heating regime ($< 6^\circ\text{C min}^{-1}$) is employed for a temperature range of around 80°C .

4.2.2. Direct 4D printed wave spring. The direct comparison of the AM fabricated and simulated wave spring (figure 9) shows that the proposed approach can accurately predict both the deployed shape as well as the mechanical properties of the direct 4D printed wave spring. By modeling the individual AM layers as plies in a laminate and controlling the material orientation and ply orientation, a complex shape with constantly changing circular AM directions can be accurately modeled. The one step simulation approach shown in figure 9(c) is particularly interesting. In the heating step, the time-temperature dependent deployment of the wave spring can be simulated. The wave spring holds the AM pre-strain and does not deform below its glass transition temperature. As soon as it reaches T_g , the wave spring deploys and stops deploying once the AM pre-strain is released. In the subsequent cooling step, the wave spring does not return to its original shape, which would be the case if the wave spring were modeled using thermal expansion. In the loading step, the wave spring's mechanical properties can be characterized without the need for a separate simulation using an updated geometry, or other workarounds such as temperature independent Young's moduli that would allow room temperature characterization even at a high simulation temperature. Due to the decoupling of the activation mode and activation magnitude in FFF direct 4D printing, the wave spring can be easily adapted to a variety of different shapes and sizes as shown by the design parameter study in figure 10. Due to the shell-based modeling approach, and thus relatively low element count, the entire study can be conducted in just under six hours. The average simulation takes around 30 s to complete on a mobile processor (Intel Core i7-10750H). This parametric study demonstrates that the modeling approach can provide designers with insights into design parameter trade-offs that may be non-intuitive, due to the need to design the structure in its fabricated state rather than activated state, thereby partially decoupling the design parameters from the final shape.

4.2.3. General implications. The proposed model represents a significant improvement in the predictive modeling of time-temperature dependent deployment and mechanical characterization of FFF direct 4D printing. In comparison to previous models, this model represents an efficient and accurate trade-off between different levels of modeling complexity. The model is more sophisticated than a typical thermal expansion model (see also section B.4 in the SI) as it can model the stiffness of the materials in addition to the deployment at various temperatures and in various time frames. This enables deployment and mechanical characterization in a single simulation without workarounds and modeling of

complex behavior such as sequential activation. The laminate modeling approach and choice of shell elements help to keep computational cost and complexity down. However, the model is less sophisticated than the more complex thermal models that simulate the heat distribution during printing to predict the AM pre-strain in the simulation [35].

Especially in sequential deployment, time-temperature dependent behavior can be crucial to accurately model the deployment of more complex direct 4D printed structures. Effects such as the stepped displacement of the bilayer between the T_g of PLA and PETG would not be accurately predicted using the common thermal-expansion-based modeling approach. In FFF direct 4D printing, the deformation mode is controlled by the AM direction and the deformation magnitude by the process parameters, such as nozzle temperature. By controlling the deformation magnitude with a modified generalized Maxwell model based on an AM pre-strain and the deformation mode with laminate modeling, the model can be applied to a variety of problems without major modifications (additional examples in figure B.4 in the SI). It can be extended to strand-based direct 4D printing by FFF by decreasing the element size to model smaller and narrower structures accordingly [9, 46, 56]. The model is also scalable for larger structures since it uses shells instead of solids. Large structures can include origami or flower patterns (see figure B.4 in the SI) and larger metastructures such as energy absorbing auxetics [57]. Since these structures consist of many thin active members, they can be modeled with a relatively low element count. A limitation is the general need for modeling abstractions and the likely dependence on custom meshing algorithms to ensure correct material orientation in large structures and correct mesh alignment if a mix of shell and solid elements is necessary.

The influences of the specific AM setup can be modeled in the AM pre-strain. This eliminates any need to modify the model for individual AM setups, although material testing and characterization is still needed. Due to the capability to model all steps accurately from deployment to mechanical characterization as well as potential further shape memory cycles in a single computationally inexpensive simulation, the approach is also uniquely suited to computational optimization problems that include direct 4D printing. The wave spring design parameter study also shows the capabilities of the proposed modeling approach for the design of direct 4D printed functional components. The material model and general modeling approach allows designers to rapidly ascertain what the complex interactions are between multiple design parameters in a direct 4D printed structure relative to the mechanical performance. It also enables this analysis at different temperatures and with complex material combinations, as shown by the multimaterial bilayer. The predictive nature of the model for complex loading such as hysteresis and long-term relaxation, multi-material structures and large deformation designs enables designers to systematically gain significant insight into the structural performance of a direct 4D printed system using a systematic method rather than the usual trial-and-error approach.

5. Conclusion

In summary, this work proposes a comprehensive modeling approach using a modified viscoelastic material model that can be used to model pre-strain in a material model for the simulation of direct 4D printed FFF structures. The presented model can accurately represent the time-temperature dependence of the initial deployment, the mechanical characterization as well as any subsequent programming of the 4D printed structures at varying temperatures. The model's performance is validated using several benchmarks including a direct 4D printed sequential bilayer consisting of PLA and PETG and a direct 4D printed wave spring. The material model is readily tunable to different AM processes and materials as the desired permanent shape is imparted via a modeled AM pre-strain in the material model. The AM pre-strain can be determined for any FFF process using a small set of fabricated benchmark samples. The model further supports a plane stress condition that allows the simulation to use shell elements, thus greatly reducing the computational cost of simulating thin structures, which are the focus of this approach. The presented modeling approach enables the design of structures that use complex direct 4D printed shape changing behavior without the drawbacks of the typically used thermal expansion workaround. It is further capable of directly including mechanical characterization and potential future shape memory cycles in a single model and simulation. This combination enables designers to gain new, quantitative information about the structural performance of direct 4D printed designs and to systematically explore the design space.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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